A Computational-Basis Metric Blind Spot under Phase-Flip Noise

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Abstract

This study asks whether the computational-basis success probability P(00)+P(11) can incorrectly certify a noisy Bell-state circuit. Starting from |Phi+> = (|00>+|11>)/sqrt(2), we apply a phase-flip channel with probability p to the first qubit and evaluate 101 deterministic density-matrix simulations for p = 0, 0.01, ..., 1. The measured success probability is exactly 1 throughout the sweep, whereas fidelity to |Phi+> falls from 1 to 0 and Re(rho[00,11]) changes from 0.5 to -0.5. At p = 0.5 the off-diagonal coherence vanishes even though the success metric remains perfect. Thus the selected task metric is blind to this phase error. This is an analytic, noiseless classical simulation of a simplified channel, not a measurement of a physical device or a complete CNOT characterization.

1. Introduction

Near-term quantum circuits are limited by imperfect gates, decoherence, and measurement noise, so a useful benchmark must state what its metric can and cannot observe [1]. A task success probability is defined by a chosen set of acceptable classical outputs. State fidelity instead asks how close the complete output state is to a target state [2]. These questions coincide only in special cases.

For a Bell-state preparation task, it is tempting to call both 00 and 11 successful and report P(00)+P(11). That metric reads diagonal density-matrix elements, but the relative phase distinguishing |Phi+> from |Phi-> is encoded off the diagonal. This study isolates that blind spot with a phase-flip channel represented by Kraus operators [3].

2. Methodology

2.1 Bell-state preparation

Order the computational basis as (|00>, |01>, |10>, |11>). Beginning with |00>, apply H to the first qubit:

```
(H x I)|00> = H|0> x I|0> = ((|0>+|1>)/sqrt(2)) x |0> = (|00>+|10>)/sqrt(2).
```

A CNOT with the first qubit as control maps |00> to |00> and |10> to |11>, giving

```
|Phi+> = (|00>+|11>)/sqrt(2) = (1/sqrt(2)) [1, 0, 0, 1]^T.
```

The density operator is the outer product:

```
rho+ = |Phi+><Phi+| = (1/2)(|00>+|11>)(<00|+<11|) = (1/2)(|00><00| + |00><11| +
|11><00| + |11><11|) [1 0 0 1] = 1/2 [0 0 0 0] [0 0 0 0] [1 0 0 1].
```

Hence rho[00,00] = rho[11,11] = 1/2 and the coherence elements rho[00,11] = rho[11,00] = 1/2.

2.2 Phase-flip channel and Kraus completeness

On the first qubit, Z1 = Z x I = diag(1, 1, -1, -1). The channel

```
E_p(rho) = (1-p) rho + p Z1 rho Z1
```

has Kraus operators E0 = sqrt(1-p) I4 and E1 = sqrt(p) Z1. Their completeness relation is

```
E0^dag E0 + E1^dag E1 = (1-p) I4^dag I4 + p Z1^dag Z1 = (1-p) I4 + p Z1^2 = (1-p) I4 + p I4 = I4.
```

Therefore the map is trace preserving. The operator-sum form also guarantees complete positivity.

Since Z1|Phi+> = (|00>-|11>)/sqrt(2) = |Phi->, the output is

```
rho(p) = (1-p)|Phi+><Phi+| + p|Phi-><Phi-| [1 0 0 1-2p] [0 0 0 0 ] = 1/2 [0 0 0 0 ] [1-2p 0 0 1 ].
```

The index-level rule is (Z1 rho Z1)[i,j] = z[i] z[j] rho[i,j], where (z[00], z[01], z[10], z[11]) = (1, 1, -1, -1). Thus:

```
rho[00,00](p) = (1-p)(1/2) + p(1)(1)(1/2) = 1/2 rho[11,11](p) = (1-p)(1/2) + p(-1)(-1)(1/2) = 1/2 rho[00,11](p) = (1-p)(1/2) + p(1)(-1)(1/2) = (1-2p)/2 rho[11,00](p) = (1-2p)/2.
```

2.3 Metrics

The computational-basis success probability reads only populations:

```
S(p) = P(00)+P(11) = rho[00,00](p)+rho[11,11](p) = 1.
```

Because the target is pure, fidelity becomes:

```
F(p) = <Phi+|rho(p)|Phi+> = (1-p)|<Phi+|Phi+>|^2 + p|<Phi+|Phi->|^2 = (1-p)(1) + p(0) = 1-p.
```

Finally, Re(rho[00,11]) = (1-2p)/2 and |rho[00,11]| = |1-2p|/2. The signed real part records the phase reversal; magnitude alone does not.

3. Experiment Setup

The implementation uses Python and NumPy for the exact density-matrix calculation, Matplotlib for normal figure generation, and pytest for the specified test suite. It evaluates p = 0, 0.01, ..., 1, writes 101 CSV rows, and renders two figures. No random sampling or quantum SDK is used.

```
python -m src.phase_error_blind_spot --output experiments/phase_error_sweep.csv --figures figures
```

The tests cover Kraus completeness, trace preservation, Hermiticity, positive semidefiniteness with an eigenvalue tolerance, analytic endpoints, and the metric blind spot.

4. Results

p	P(00)+P(11)	Fidelity	Re(rho[00,11])	|rho[00,11]|
0.00	1.00	1.00	0.50	0.50
0.25	1.00	0.75	0.25	0.25
0.50	1.00	0.50	0.00	0.00
0.75	1.00	0.25	-0.25	0.25
1.00	1.00	0.00	-0.50	0.50

Representative values from the deterministic 101-point sweep. The task metric remains perfect while fidelity and signed coherence expose the phase error.

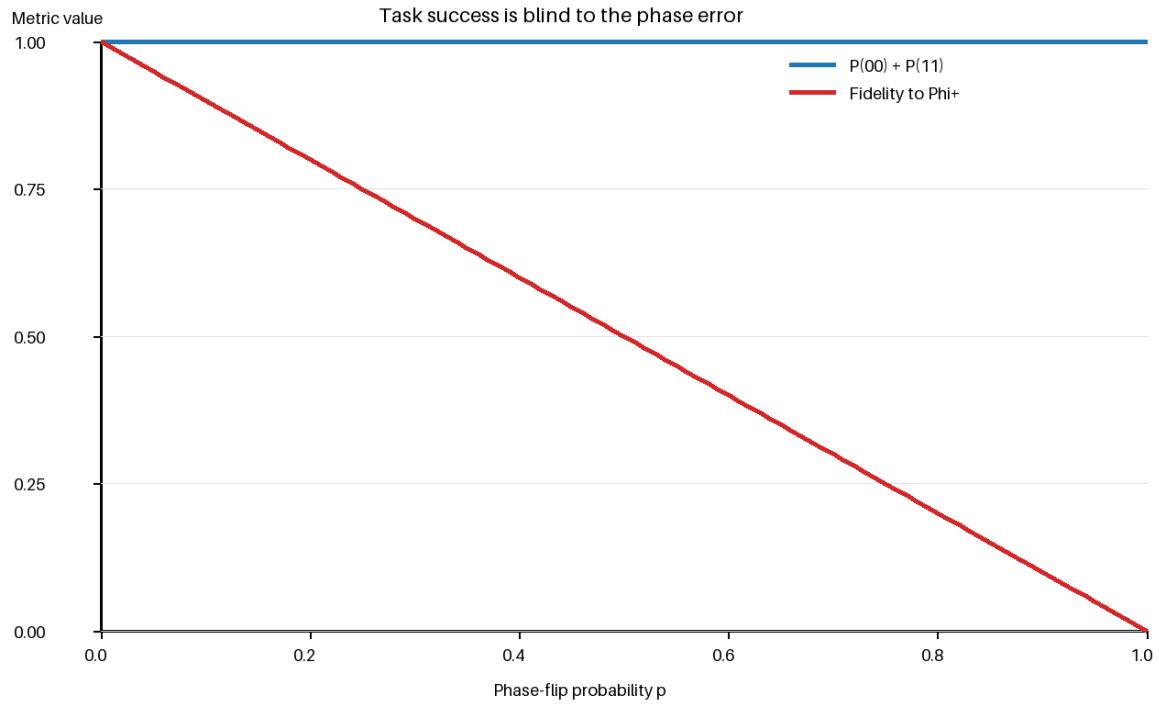


Figure 1. Computational-basis success remains one for every phase-flip probability, while fidelity falls linearly from one to zero. The chosen task metric therefore cannot certify the intended Bell state.

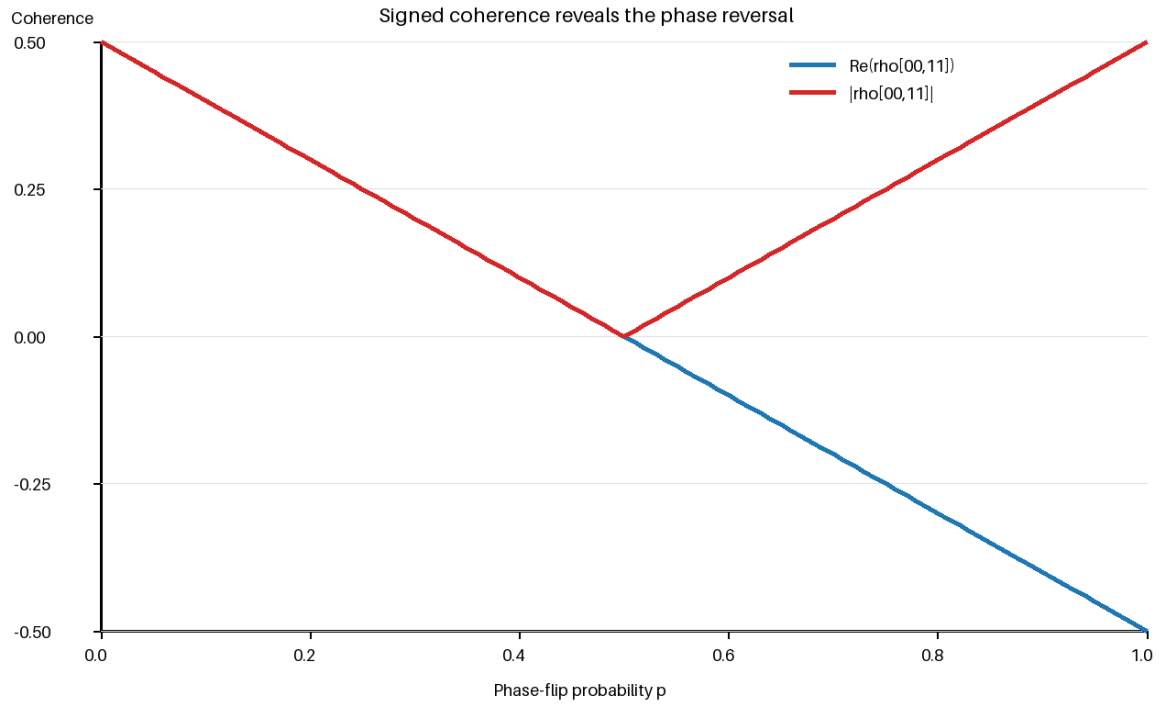


Figure 2. The real coherence crosses zero at p = 0.5 and becomes negative, whereas its magnitude returns to 0.5. At p = 1 the state is the coherent but orthogonal |Phi->, not the incoherent midpoint mixture.

At p = 0.5, rho(0.5) = (1/2)|00><00| + (1/2)|11><11|. It has no |00><11| coherence but still has S = 1. At p = 1, rho(1) = |Phi-><Phi-|: coherence magnitude is restored, its sign is reversed, fidelity to |Phi+> is zero, and S is still one.

5. Conclusion

Yes. Computational-basis success probability alone can judge this noisy Bell-state circuit as perfectly successful after its coherence has vanished or its relative phase has reversed. In this model S(p) = 1 for all p, while F(p) = 1-p and Re(rho[00,11]) = (1-2p)/2. A benchmark intended to certify Bell-state preparation must therefore include phase-sensitive information rather than only accepted computational-basis outcomes.

6. Discussion

This study is deliberately narrow. It is an analytic density-matrix result reproduced by deterministic classical simulation, not hardware data. A phase-flip channel is a simplified model, and no SPAM error, crosstalk, coherent calibration error, finite-shot uncertainty, or drift is included. The channel is applied after an ideal Bell circuit rather than assigned to a specific pulse or time location inside a physical CNOT implementation. The results do not estimate the CNOT error rate of an IBM device or any other processor, and they do not constitute randomized benchmarking or full CNOT characterization.

Fidelity to one target answers whether the state matches |Phi+>; it is not a complete description of a general quantum process. Coherence magnitude likewise loses the sign that distinguishes |Phi+> from |Phi->. The comparison is useful precisely because the metrics expose different projections of the same state.

References

[1] J. Preskill, 'Quantum Computing in the NISQ Era and Beyond,' Quantum 2, 79 (2018). DOI: 10.22331/q-2018-08-06-79.

[2] M. A. Nielsen and I. L. Chuang, Quantum Computation and Quantum Information, 10th Anniversary Edition, Cambridge University Press (2010). DOI: 10.1017/CBO9780511976667.

[3] C. J. Wood, J. D. Biamonte, and D. G. Cory, 'Tensor Networks and Graphical Calculus for Open Quantum Systems,' Quantum Information and Computation 15, 759-811 (2015), arXiv:1111.6950.